Project Topic:	**CSIT-26-S2-05: A Mobile Application for Wise Workout**		
Project Group:	**FYP-26-S2-37**		
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Project Description:

This project aims to develop a mobile application (for Android and iOS) that can collect exercise data from cell phone sensors or wearable devices, provide training and fitness advice, and facilitate social interactions such as initiating competitions between registered users. The application should have the following features.

1. *Collect exercise data:*
The application can collect user exercise information from either the cell phone's own sensors or from smart wearable devices (via Wi-Fi, Bluetooth, etc.).

2. *Estimate exercise effects* (short-term or long-term) and provide analysis:
The application can estimate the exercise results of each activity, a day, a week, a month, or any period, such as the number of calories burned and the average and maximum heart rate.

3. *Supply fitness advice and schedule a fitness plan*:
The application can provide fitness advice according to the user's fitness goals and their entered health information. It can also create a fitness plan based on the fitness goal, and users can customize the plan to their preferences, such as the length of the plan or the daily weight loss goal.

4. *Remind user of exercise or take a break*:
The application can send reminders to users to exercise if they have not met their scheduled exercise goals. It can also send reminders to users to take a break if they have been exercising too much.
Team to develop a product based on a suitable business model with responsive and user-friendly interface that works seamlessly on various mobile devices. Optimize the user experience for different screen sizes.

5.Connect with social media and initiate competitions between users:
The application can be used to share exercise completion data on social media platforms such as Facebook, Instagram, Twitter. TikTok. Users can also initiate competitions with other registered users and invite them to join in. The competition results and rankings can be shared on social media.

Team to develop a *SaaS* product based on suitable business model with above functionalities and offer a user-friendly solution that provides profile-based services to the users. The product must empower users to make data-driven decisions based on their profile and the activities they carry out.

The application must have features to profile the users, Intelligent content delivery with context aware recommendations, Incentive based motivation mechanism with gamification to facilitate social interaction, Analytics- including Progress tracking, predictive analytics etc.

The application must have product marketing website to market and attract the users to sign on and then download the mobile application.

The team may consider releasing the application *to the app store*.

Project Schedule
Released on: 4 April 2026 Rev 0

Tasks	**Target Date**	**Activities and Deliveries**
(Session 1) 4 Apr 2026 – 22 Jun 2026 **Project Briefing**	4 Apr 2026	Start of Final Year Project. Project members attending the 3-hour lectures
Forming the team and setting up project website	11 Apr 2026	• The students are to form a project team and identify a team leader. The project team to share the team members' profile and Leader details. • . • To set up project website. All project meetings need to be minuted and posted in the project website. Blog must be accessible to Supervisor and Assessor • Carry out preliminary research in topic area individually **Deliverable: The team members and team leader details, Website link, Reflective Diray-1. (11 Apr 2026)** **Reflective Diary needs to submit every week, individually- Start from 11 Apr 2026**
Research on topic of project	Week 1 – 4: 4 Apr 2026 – 2 May 2026	• To do a literature review and submit a Project requirement Documentation (PRD) by 2nd May 2026). • You are required to explore the literature to find answers to the following questions: ○ What is the different work out monitoring applications available in the market?

Tasks	**Target Date**	**Activities and Deliveries**
		○ What features do they offer? ○ Do they provide profile-based recommendations to users? ○ What are the important functionalities of such app? ○ How do they collect data from the devices? ○ What are the common protocols used for such data acquisition? ○ How is such app implemented? ○ How do they link to social networks to encourage participation and competition? ○ What type of business model is suitable for such implementations? ○ What type of alert system do you intend to provide to the users? ○ Do they provide cross platform features without compromising user experience? ○ What are the common software tools available to implement such systems? ○ What are the functionalities you intend to build in your product? ○ Who are the target users? ○ How do you plan to implement your project? What algorithm and how to apply your algorithm / technique? What are the recent developments in the topic area? ○ Do you have any case study to support your case? ○ How do you model such a system? **Deliverable: Project Requirement Documentation contains Research Summary, Functional Hierarchy diagram, Use case diagrams, Project Schedule, Development tools, development methodologies (2nd May 2026)** Submit in Moodle by 9th May
Requirement	Week 5 - 7 3 May 2026 – 24 May 2026	Gather information on the requirements of the project. • The project team is to identify the system's services (functionalities), constraints and goals of the project. ○ Identifying the various functionalities, algorithms as well as interfaces required from three perspectives - functional, non-functional, and security. The team is to produce Refined Project functionalities and System Requirement Specification. **Deliverable: System Requirement Specification (14th May 2026)**
Analysis and design	Week 8-9: 25 May 2026-7 Jun 2026	Project team to start with solutions for the implementation of the project.

Tasks	**Target Date**	**Activities and Deliveries**
		• The team is to produce the System Designs, Architectural Designs, Database Design of the system, wireframes etc. • Use suitable modelling and visualization tools • Finalize functionalities and start work on the prototype **Deliverable: Technical Design Manual with functional prototype (21 May 2026)**
Implementation	Week 10-11 8 Jun 2026 21 Jun 2026	Depending on the systems development methodology used by the team, to continue the development of the project. Complete the development of prototypes with basic functionalities defined for the system design manual. Complete the Progress report with preliminary documents and submit it to supervisor. **Deliverable: System prototype with basic functionalities, Project Progress Report consists of the following: Preliminary Project Documents** (Preliminary Technical Documents, Preliminary User Manual, Peer Assessment form for review by Supervisor **(10 Jun 2026)** ***Submit in Moodle by 13 June***
End of Term 1 Review	Week 11 20 Jun 2026	• Demonstrate the prototypes to the supervisor and assessor **Deliverable: Project Progress Report (Preliminary Technical Documentation, Preliminary User Manual, Application Prototype with basic functionalities (20 Jun 2026).**
<u>(Term 2) 22 Jun – 2026- 23 Aug 2026</u> **Implementation**	Week 1 – 3: 21 Jun 2026- 11 Jul 2026	• Continue to work on the development. • As the team continues to develop, there might be changes to the original technical design. Update your Technical Design Documents. • Demonstrate functionalities at module level • The team to produce a module System Test Plan. • Carry out Module Testing **Deliverable: Individual Functional Modules, Centralized DB, Module Test Plan. (11 Jul 2026).**
	Week 4– 6: 12 Jul 2026 – 1 Aug 2026	• The team is to update the technical documentation and user Manual • Continue with development of all functional modules and test in an integrated environment • Demonstrate Completed modules with full functionalities in an integrated environment • The team is to carry out the Integration Test **Deliverable: Integrated Module functionalities, and Progress Report with final document- Draft, User**

Tasks	Target Date	Activities and Deliveries
		Manual- Draft, Integration Test Summary (1st Aug 2026).
Prepare for Project Presentation and Demonstration of product	Week 7 – 8 2 Aug 2026 15 Aug 2026	• The team is to wrap up the development and prepare for the project presentation. Create Project Video • To present the final fully functional product and the documents to the project supervisor. **Deliverable: The Final Project Documentations- Technical Documentation, User Manual. Final fully functional Product, Project Video, Presentation slides.** **Demonstrate to the Supervisor on 13th Aug 2026** **Submit the deliverables in Module on 15 August**
Project Final Presentation	Week -9-10 16 Aug 2026 22 Aug 2026	• **The team is to present and demonstrate the final product to the supervisors and an Assessor.** **Deliverable: Final Project Documentations, including Technical Documentation, User Manual, Project Video, Peer Assessment form, Source Code. Final, fully functional Product Demonstration to** **Assessor and Supervisors. (22 Aug 2026)**